2025 Final

$\boxed{1}$.

(a) $y = X\beta + e$

$$= (1 \quad X_1)\begin{pmatrix}\beta_0 \\ \beta_1^T\end{pmatrix} + e$$

$$= 1\beta_0 + X_1\beta_1^T + e, \qquad 1\alpha = 1\beta_0 + 1\beta_1^T\bar{x}$$

$$= 1\alpha - 1\beta_1^T\bar{x} + X_1\beta_1^T + e$$

$$= 1\alpha + (I - \tfrac{1}{n}J)X_1\beta_1 + e$$

$$= 1\alpha + X_c\beta_1 + e$$

$$= (1 \quad X_c)\begin{pmatrix}\alpha \\ \beta_1\end{pmatrix} + e$$

$$= X^*\begin{pmatrix}\alpha \\ \beta_1\end{pmatrix} + e$$

$\blacksquare$

(b) $C_X = C_1 \oplus C_{(I-H_1)X_1}$

$\Rightarrow H_X = H_1 + H_{X_c}$

$SSR = y^T(H_X - H_1)y$

$$= y^T H_{X_c} y$$

$\blacksquare$

(c) $R^2 = \frac{SSR}{SST} = \frac{y^T(H_x - H_1)y}{\underbrace{y^T(I-H_1)y}_{= S_{yy}}}$

WTS: $y^T(H_x - H_1)y = S_{yx}^T S_{xx}^{-1} S_{yx}$

$$
\begin{aligned}
y^T(H_x - H_1)y &= y^T H_{x_c} y \\
&= y^T X_c (X_c^T X_c)^{-1} X_c^T y \\
&= y^T X_c S_{xx}^{-1} X_c^T y \\
&= y^T (I - H_1) x\, S_{xx}^{-1} x^T (I - H_1) y \\
&= S_{yx}^T S_{xx}^{-1} S_{yx}
\end{aligned}
$$

$\because\ X_c^T y = x(I - \frac{1}{n}J)y = S_{yx}$

$\boxtimes$

$\boxed{2}$

(a) $\begin{pmatrix} y \\ x_1 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} y \\ x^T \end{pmatrix} \sim N_{p+1}(\mu, \Sigma)$

$\mu = (\mu_y, \mu_x^T)^T, \quad \Sigma = \begin{pmatrix} \sigma_{yy} & \sigma_{yx}^T \\ \sigma_{yx} & \Sigma_{xx} \end{pmatrix}$

$$\frac{f(y,x)}{f(x)} = \frac{\dfrac{1}{(2\pi)^{\frac{p+1}{2}} |\Sigma|^{1/2}} e^{-\frac{1}{2}((y,x^T)-\mu)^T \Sigma^{-1} ((y,x^T)-\mu)}}{\dfrac{1}{(2\pi)^{p/2} |\Sigma_{xx}|^{1/2}} e^{-\frac{1}{2}(x-\mu_x)^T \Sigma_{xx}^{-1} (x-\mu_x)}}$$

$$|\Sigma| = |\Sigma_{xx}| \, |\sigma_{yy} - \sigma_{yx}^T \Sigma_{xx}^{-1} \sigma_{yx}|$$

$$(y-\mu_y, x-\mu_x)^T \Sigma^{-1} \begin{pmatrix} y-\mu_y \\ x-\mu_x \end{pmatrix}$$

$S := \sigma_{yy} - \sigma_{yx}^T \Sigma_{xx}^{-1} \sigma_{yx}$

$$\Sigma^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}\sigma_{yx}^T \Sigma_{xx}^{-1} \\ -\Sigma_{xx}^{-1}\sigma_{yx} S^{-1} & \Sigma_{xx}^{-1} + \Sigma_{xx}^{-1}\sigma_{yx} S^{-1} \sigma_{yx}^T \Sigma_{xx}^{-1} \end{pmatrix}$$

$$(y-\mu_y)^T S^{-1}(y-\mu_y)$$

$$-(y-\mu_y)^T S^{-1}\sigma_{yx}^T \Sigma_{xx}^{-1}(x-\mu_x)$$

$$-(x-\mu_x)^T \Sigma_{xx}^{-1}\sigma_{yx} S^{-1}(y-\mu_y)$$

$$+\boxed{(x-\mu_x)^T \Sigma_{xx}^{-1}(x-\mu_x)} \quad \mathrm{I}-\mathrm{I}=0$$

$$+(x-\mu_x)^T \Sigma_{xx}^{-1}\sigma_{yx} S^{-1}\sigma_{yx}^T \Sigma_{xx}^{-1}(x-\mu_x)$$

$$\Rightarrow \left((y-\mu_y)-\sigma_{yx}^T\Sigma_{xx}^{-1}(x-\mu_x)\right)^T S^{-1}\left((y-\mu_y)-\sigma_{yx}^T\Sigma_{xx}^{-1}(x-\mu_x)\right)$$

$$f(y|x) = \frac{1}{(2\pi)^{1/2}\left|\sigma_{yy}-\sigma_{yx}^T\Sigma_{xx}^{-1}\sigma_{yx}\right|^{1/2}}$$

$$\times\, e^{-\frac{1}{2}\left((y-\mu_y)-\sigma_{yx}^T\Sigma_{xx}^{-1}(x-\mu_x)\right)^T S^{-1}\left((y-\mu_y)-\sigma_{yx}^T\Sigma_{xx}^{-1}(x-\mu_x)\right)}$$

$$\mathbb{E}[y|x] = \mu_y + \sigma_{yx}^T\Sigma_{xx}^{-1}(x-\mu_x) = \beta_0+\beta_1 x$$

$$\beta_0 = \mu_y - \sigma_{yx}^T\Sigma_{xx}^{-1}\mu_x$$

$$\beta_1 = \sigma_{yx}^T\Sigma_{xx}^{-1}$$

$$\mathrm{Var}(y|x) = \sigma_{yy} - \sigma_{yx}^T\Sigma_{xx}^{-1}\sigma_{yx}$$

$\blacksquare$

(b)

$$f(y,x)=\frac{1}{(2\pi)^{\frac{p+1}{2}}|\Sigma|^{1/2}}e^{-\frac{1}{2}((y,x^T)-\mu)^T\Sigma^{-1}((y,x^T)-\mu)}$$

$$L(\mu,\Sigma)=\frac{1}{(2\pi)^{\frac{n(p+1)}{2}}|\Sigma|^{n/2}}e^{-\frac{\sum_{i=1}^{n}((y,x^T)-\mu)\Sigma^{-1}((y,x^T)-\mu)}{2}}$$

$$\ell(\mu,\Sigma)=-\frac{n(p+1)}{2}\log 2\pi-\frac{n}{2}\log|\Sigma|-\frac{\sum_{i=1}^{n}((y,x^T)-\mu)^T\Sigma^{-1}((y,x^T)-\mu)}{2}$$

$$=-\frac{n(p+1)}{2}\log 2\pi-\frac{n}{2}\log|\Sigma|-\frac{1}{2}tr\left(\Sigma^{-1}\underbrace{\sum_{i=1}^{n}((y,x^T)-\mu)^T((y,x^T)-\mu)}_{=nS}\right)$$

$$\frac{\partial \ell}{\partial \mu}=\sum_{i=1}^{n}\Sigma^{-1}((y,x^T)-\mu)=0$$

$$\Rightarrow \sum_{i=1}^{n}(y,x^T)-n\mu=0$$

$$\Rightarrow \hat{\mu}=(\bar{y},\bar{x}^T)$$

$$\frac{\partial l}{\partial \Sigma} = -\frac{n}{2}\Sigma^{-1} + \frac{n}{2}\Sigma^{-1} S \Sigma^{-1} = 0$$

$$\Rightarrow -\frac{n}{2}\Sigma + \frac{n}{2} S = 0$$

$$\Rightarrow \hat{\Sigma} = S$$

$$\Rightarrow \widehat{\Sigma^{-1}} = S^{-1}$$

$$\hat{\beta}_0 = \bar{y} - S_{yx}^T S_{xx}^{-1} \bar{x}$$

$$\hat{\beta}_1 = S_{yx}^T S_{xx}^{-1} = S_{xx}^{-1} S_{xy}$$

(c) $SST = y^T(I - \frac{1}{n}J)y$

$$\hat{y} = \hat{\beta}_0 + x\hat{\beta}_1 = \bar{y} - \bar{x} S_{xx}^{-1} S_{xy} + x S_{xx}^{-1} S_{xy}$$

$$= \bar{y} + (x - \bar{x}) S_{xy}^{-1} S_{xy}$$

$$(x - \bar{x}) S_{xx}^{-1} S_{xy} = \hat{y} - \bar{y} = (H_x - H_1)y$$

$$(I - H_1)x\{x^T(I - H_1)x\}^{-1} x^T (I - H_1) y = (H_x - H_1)y$$

$$\Rightarrow H_{x_c} y = (H_x - H_1)y$$

$$y^T(H_x - H_1)y = y^T(H_x - I + I - H_1)y$$

$$y^T H_{x_c} y = y^T(I - H_1)y - y^T(I - H_x)y$$

$$\Rightarrow \underbrace{y^T(I - H_1)y}_{SST} = \underbrace{y^T(I - H_x)y}_{SSE} + \underbrace{y^T H_{x_c} y}_{SSR} \quad \text{still holds}$$

∎

(d) $H_1 \hat{y} = H_1 H_x y = H_1 y$

$$\left(\frac{(y-\bar{y})^T(\hat{y}-\bar{y})}{\sqrt{y^T(I-H_1)y}\sqrt{y^T(H_x-H_1)y}} \right)^2 = \left(\frac{y^T(I-H_1)(H_x-H_1)y}{\sqrt{y^T(I-H_1)y}\sqrt{y^T(H_x-H_1)y}} \right)^2$$

$$= \frac{y^T(H_x-H_1)y}{y^T(I-H_1)y}$$

$$= \frac{SSR}{SST} = R^2$$

$\boxtimes$

$\boxed{3}$.

$$\overline{y_{i\cdot}} = \mu + a_i + \overline{\varepsilon_{i\cdot}}$$

$$\mathbb{E}\,\overline{y_{i\cdot}} = \mu$$

$$\begin{aligned} \operatorname{var}(\overline{y_{i\cdot}}) &= \operatorname{var}(a_i) + \operatorname{var}(\overline{\varepsilon_{i\cdot}}) \\ &= \sigma_a^2 + \frac{\sigma^2}{n_i} \\ &= A_i \end{aligned}$$

$$\begin{pmatrix} \overline{y_{1\cdot}} \\ \overline{y_{2\cdot}} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \mu \end{pmatrix}, \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} \right)$$

$$\hat{\mu} = (\mathbf{1}^T V^{-1} \mathbf{1})^{-1} \mathbf{1}^T V^{-1} y$$

$$= \frac{\frac{1}{A_1}\overline{y_{1\cdot}} + \frac{1}{A_2}\overline{y_{2\cdot}}}{\frac{1}{A_1} + \frac{1}{A_2}}$$

$$\frac{1}{A_1} + \frac{1}{A_2} = \frac{A_1 + A_2}{A_1 A_2}$$

$$= \frac{A_2}{A_1 + A_2}\overline{y_{1\cdot}} + \frac{A_1}{A_1 + A_2}\overline{y_{2\cdot}}$$

$$\hat{a}_i^{BLUP} = \underbrace{E[a_i]}_{\text{BLUE}} + cov(a_i, \bar{y}_{i\cdot})\, var(\bar{y}_{i\cdot})^{-1} (\bar{y}_{i\cdot} - \hat{\mu}^{BLUE})$$

$$= 0 + \sigma_a^2 \left(\frac{1}{\sigma_a^2 + \frac{\sigma^2}{n}}\right)(\bar{y}_{i\cdot} - \hat{\mu})$$

$$= \frac{\sigma_a^2}{A_i}(\bar{y}_{i\cdot} - \hat{\mu})$$

$\boxtimes$

$\boxed{4}$.

$$l_c(\beta,\sigma^2,\mu_x,\tau^2) = \sum_{i=1}^{n}\log f(x_i;\mu_x,\tau^2) + \sum_{i=1}^{n}\log f(y|x_i;\beta,\sigma^2)$$

$$l_c = -\frac{n}{2}\log\tau^2 - \frac{1}{2\tau^2}\sum_{i=1}^{n}(x_i-\mu_x)^2 - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\beta x_i)^2$$

$$\theta^{(t)} = (\beta^{(t)}, \sigma^{2(t)}, \mu_x^{(t)}, \tau^{2(t)})$$

$$y_i|x_i,\theta^{(t)} \sim N(\beta^{(t)}x_i, \sigma^{2(t)})$$

$$\mathbb{E}^{(t)}(y_i) = \beta^{(t)}x_i$$

$$\mathbb{E}^{(t)}(y_i^2) = \sigma^{2(t)} + (\beta^{(t)}x_i)^2$$

$$\mathbb{E}^{(t)}[(y_i-\beta x_i)^2]$$

$$= \mathbb{E}^{(t)}(y_i^2) - 2\beta x_i\mathbb{E}^{(t)}(y_i) + \beta^2x_i^2$$

$$= \sigma^{2(t)} + (\beta^{(t)}x_i)^2 - 2\beta x_i\beta^{(t)}x_i + \beta^2x_i^2$$

$$= \sigma^{2(t)} + (\beta^{(t)}x_i - \beta x_i)^2$$

$$= \sigma^{2(t)} + (\beta^{(t)} - \beta)^2x_i^2$$

$$Q(\theta|\theta^{(t)}) = -\frac{n}{2}\log\tau^2 - \frac{1}{2\tau^2}\sum_{i=1}^{n}(x_i-\mu_x)^2$$
$$-\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\left[\sum_{i=1}^{k}(y_i-\beta x_i)^2 + \sum_{i=k+1}^{n}\{\sigma^{2(t)} + (\beta^{(t)}-\beta)^2x_i^2\}\right]$$

$\mu_x^{(t+1)} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n}x_i$ by $\frac{\partial Q}{\partial\mu_x}$

$\tau^{2(t+1)} = \frac{1}{n}\sum(x_i-\bar{x})^2$ by $\frac{\partial Q}{\partial\tau^2}$

$$\sum_{i=1}^{k}(y_i-\beta x_i)^2+\sum_{i=k+1}^{n}(\beta^{(t)}-\beta)^2 x_i^2$$

$$\frac{\partial Q}{\partial \beta} \Longrightarrow -2\sum_{i=1}^{k}x_i(y_i-\beta x_i)-2\sum_{i=k+1}^{n}x_i^2(\beta^{(t)}-\beta)=0$$

$$\Rightarrow \sum_{i=1}^{n}\beta x_i^2=\sum_{i=1}^{k}x_i y_i+\beta^{(t)}\sum_{i=k+1}^{n}x_i^2$$

$$\Rightarrow \beta^{(t+1)}=\frac{\sum_{i=1}^{k}x_i y_i+\beta^{(t)}\sum_{i=k+1}^{n}x_i^2}{\sum_{i=1}^{n}x_i^2}$$

$\boxtimes$